PathFinder: An Indoor Navigation Web Application for the Physical Sciences Building in the University of the Philippines Los Baños

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***Abstract*— Indoor wayfinding remains challenging in multi-story buildings, where satellite-based positioning is unreliable and indoor positioning systems require costly, hardware-dependent infrastructure that is difficult to scale across floors and buildings. This study developed and evaluated a web-based indoor navigation application for the Physical Sciences Building at the University of the Philippines Los Baños, designed as a scalable and software-driven alternative to hardware-based positioning and static physical floor plans. The application provides a navigation interface that guides users between selected origin and destination points, and a layout builder that allows building administrators to create and publish digital map layouts for display within the application. Usability was evaluated with two participant groups: six regular users, who assessed navigation accuracy through a standardized usability questionnaire and open-ended questions, and four building administrators, who assessed the layout builder's functionality qualitatively. Users reported a mean usability score of 81.67 out of 100, rated "Excellent," with positive feedback on route clarity, while administrators found the layout builder intuitive but identified specific areas for refinement. These findings support the application as a viable, scalable approach to indoor navigation.**

***Index Terms*— Indoor Navigation, Wayfinding, Layout Builder, Web Application**

I. Introduction

A. Background of the Study

The world is a huge place, and we may often find ourselves getting lost in unfamiliar places. Throughout history, we have invented various tools starting with the compass and the map to help us with navigation. Nowadays, navigation is made easy thanks to modern tracking and mapping technology. In the Philippines, applications like Waze help to ease the transportation crisis [1].

One of the most commonly used technologies is the GPS (global positioning system), a publicly available mapping system created by the US government that uses satellites to approximate the location of a GPS receiver contained in a device such as a laptop or a mobile phone. This allows users to obtain directions to and from a particular place, and it is especially useful in a travel setting. Popular applications that use GPS for navigation include Google Maps, Waze, and Apple Maps.

However, due to its top-down view from the satellites, GPS technology only works best outdoors. It does not work well for buildings with multiple levels. To solve this, we can use an indoor positioning system (IPS) which works like a miniature version of the GPS. Specifically, we can install specialized tracking devices in various places inside the building. Several technologies can be used as positioning methods for the IPS, some of which include Wi-Fi, infrared, Bluetooth, and computer vision.

To further improve the IPS in terms of visualization, we can implement an indoor map where we can mark the locations of each landmark, as well as the location of the user. These maps can be derived from the floor plans of the building, and they can be implemented without an IPS. Examples of applications that allow us to design indoor maps are Mappedin and ArcGIS Indoors. Google Maps also supports indoor maps, but only a few buildings currently have this functionality.

B. Problem Statement

Globally, hundreds of thousands of buildings are being constructed every year. In the Philippines, the total number of buildings constructed in the year 2022 was 169,432, which was 4.2 percent higher than 162,619 constructions in 2021 [2]. Since GPS is not an effective solution for mapping the inside of these buildings, this means that there is a growing need for indoor mapping solutions.

The Physical Sciences building in the University of the Philippines Los Baños campus is one of these buildings that need indoor mapping. New students and visitors may find it difficult to navigate inside this building due to its multiple floors and classrooms. Currently there is a floor plan located in this building, but it is not easy to find.

The IPS is a possible solution for mapping the Physical Sciences building, but it can be costly and challenging to build and maintain. It is also not very scalable if we would expand the system to the entire campus because it requires external hardware for each floor in each building. Hence, we turn to indoor mapping apps that can be accessed anywhere.

C. Objectives of the Study

The main objective of this study is to develop a web application that provides an indoor map of the UPLB Physical Sciences Building. Specifically, it aims to:

1.) Integrate point to point directions to aid users in finding the exact locations of classrooms, offices, and facilities within the UPLB Physical Sciences Building.

Presented to the Faculty of the Institute of Computer Science, University of the Philippines Los Baños in partial fulfillment of the requirements for the Degree of Bachelor of Science in Computer Science

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2.) Enable building administrators to generate a building's maps based on floor plans, allowing for updates and customization of the mapping content.

3.) Assess the usability of the system through user testing to evaluate navigation accuracy, interface intuitiveness, and overall user satisfaction.

D. Significance of the Study

The proposed web application will be beneficial to visitors and new students who have classes inside the Physical Sciences building. It will be easier for them to find the classrooms that they need to get to, and it will also save them time and prevent traffic in the corridors.

The results of this study may also serve as a basis for future indoor mapping implementations in UPLB.

Aside from determining the feasibility of such a project, future studies will know what can be improved based on this study.

E. Scope and Limitations of the Study

The main focus of this research is to build a web application to navigate through the inside of the Physical Sciences building located at the University of the Philippines Los Baños, Laguna. The end-users of this research include individuals who are unfamiliar with the building, regular occupants of the building, such as employees and students, and building administrators of the Physical Sciences building. Using the application requires access to network connectivity, as the application may need to update map data and navigation information as it is being used, and it is also required when logging in.

II. Review of Related Literature

A. Technological Trends in Mapping Applications

Location-based services (LBS) are an umbrella term for technologies that use a device's location to provide services. Object positioning, navigation, and mapping all count as LBS. [3] observed the difference between LBS and non-LBS systems through an empirical study involving navigation tasks, and it was found that individuals prefer LBS because it is efficient and easy to use. However, these individuals also had privacy concerns, and they were willing to provide their location to network providers but hesitant to give the same information to other entities such as third-party apps.

Another technology that is closely related to LBS is the Geographic Information System (GIS). GIS is similar to LBS in that they both involve spatial and location data, but one main difference lies in their intended purpose: GIS is often used for mapping and analyzing geographic information, while LBS is more user-oriented and focuses on navigation and adding that same functionality in everyday apps. GIS has made great progress over the past decades, but there are still areas for improvement, such as geographic modeling enhancement and human-oriented presentation of geographic information [4].

LBS and GIS are far from the only technological trends. [5] provides an overview of various indoor positioning and wayfinding methods from 2014 to 2020. Among these technologies are radio frequency identification (RFID), ultra-wideband (UWB), and computer vision-based indoor mapping.

B. Commonly Used Navigation Apps

To understand how navigation technologies work, we first take a look at Google Maps, one of the most popular and feature-complete navigation apps. A report by [6] analyzes this app in terms of its navigation and mapping. In terms of navigation, Google Maps uses the A* pathfinding algorithm. For its mapping, Google uses specialized devices such as a 360° camera attached to a car. This improves the accuracy of paths and provides 3D viewing through Street View. To calculate the estimated time of arrival (ETA), Google Maps collects data from its users on that particular route and approximates the traffic speed. This means that the more users there are, the more accurate the estimates will be. However, there are also ethical considerations here, and transparency is important so that the user knows if their personal information is not being used against them. Aside from Google Maps, an alternative mapping solution is OpenStreetMap, a community-driven digital map where volunteers can contribute to its development [7]. OpenStreetMap emphasizes collaboration, organizing 'mapping parties' that annotate data on various geographical areas.

C. Campus Navigation Solutions

There exist several studies that involve indoor mapping for universities around the world. [8] implemented a navigation app for Jhulelal Institute of Technology in Nagpur, India. The application was created to help college students reach JIT College from various outside places (inter-college), as well as offer directions to buildings and classrooms within the university (intra-college). For the inter-college stage, they used a GPS and the Apple Maps API with a fixed destination towards JIT College. For the intra-college stage, the paths to and from each building were hard-coded, and a pre-made image was shown on the screen. This method works, but it is not ideal since the number of possible paths is close to n2 (where n is the number of buildings), so having a large number of buildings increases the manual work exponentially.

There are also several navigation studies in the Philippines. [9] created a mobile navigation app for the Technological University of the Philippines (TIP) Quezon City campus, and it is intended for new students and visitors. First, the user is taken to a login screen. After creating an account and logging in, the user can choose a source and a destination building, and the app generates a path between them using a fuzzy logic algorithm. The advantage of this feature over [8] is that the paths are automatically generated. However, it may be prone to errors depending on the strength of the algorithm. As an added feature, the user can also see other active users (shown as pinpoints) within the same area using a channel selection algorithm through Bluetooth.

Similar to [9], [10] created a locator map of Saint Michael's College of Laguna using Android Studio. It uses the A* algorithm for pathfinding. The layout of the campus map (including the buildings and the walkable roads) was manually created. The app also has a search bar where the user can search for a particular building, and the app also displays information about that building, including the classes one

needs to take in that building. Overall, the app was marked as "Excellent" based on 346 respondents.

[11] had a more gamified approach to campus navigation. The mobile app for the Don Mariano Marcos Memorial State University campus map incentivizes the user to go to different locations within the campus to collect BeeCoins which represent the number of points the user has. Through this approach, the user can have fun while learning. The app used satellite view for the map layout and GPS for the user's location.

D. Indoor Navigation in Other Industries

Aside from campus navigation, indoor mapping can also be effective in other settings like finding stores in malls. A proposed system by [13] uses Wi-Fi fingerprinting to locate the user and Dijkstra's algorithm for pathfinding. The application was developed using Android Studio and Node.js, and it can display directions from the user's location to other stores in the mall. However, it is unclear how a multi-floor navigation functionality can be achieved based on this implementation.

Another implementation by [14] combines augmented reality (AR), geographic information systems (GIS), and GPS technologies to find stores. To do this, a database was kept storing data on the floor level, latitude, and longitude of each store. Then, the user could use their phone camera to locate the store through an AR popup. Again, it is unclear whether the floor level of the user was automatically determined or manually inputted.

Similar to [13], [15] uses Wi-Fi instead of GPS to improve the accuracy of the user's indoor location. The mobile application used an existing indoor map found in Google Maps and provided an interface for locating stores inside Plaza Low Yat in Kuala Lumpur, Malaysia. On a scale of 1-5, users consistently rated it above 4 in terms of its usefulness, ease of use, efficiency, and aesthetics.

E. Challenges of Indoor Navigation

While indoor navigation is certainly beneficial to all kinds of users, it does not come without its challenges. A study by [16] proposes a set of indoor mapping and modeling (IMM) issues that are worth researching. Among these problems are optimal navigation routing, emergency preparedness, and legal and security issues.

Indoor navigation is believed to promote accessibility features, especially for users with mobility challenges. However, there is still a lack of actual testing on these individuals. According to an analysis of 111 existing wayfinding and navigation papers [17], it was found that in most papers, persons with mobility issues have had no involvement with the design and development of navigation solutions. This makes it hard to prove the effectiveness of the product in real-life contexts.

Another problem with indoor navigation is its lack of accessibility standards and guidelines. [18] reports that in OpenStreetMap, on average, only one building per city has an indoor map, and among those buildings, the ones with accessibility information are even smaller.

One of the biggest challenges of indoor positioning, mapping, and navigation (IPMN) is its lack of standardization. Over the decades there have been multiple researchers and developers tackling IPMN, but it is hard to undergo the process of industrialization because there is no clear set of rules to be followed. [19] also points out the lack of a standardized file format for indoor map data, making it harder to have a one-size-fits-all approach to speed up the indoor mapping process globally. To solve this, [20] have proposed a set of requirements for IPMN: 1.) Indoor positioning can be classified into two types: requiring external equipment, or device-free. If using the former, the external equipment must be prepared in advance; 2.) Indoor positioning must not be separated from map visualization, and map visualization should consider expressing symbols, color, and semantic information; 3.) Indoor and outdoor positioning should be treated as separate modes in the application; and 4.) A consistency test is required for evaluating the accuracy and cost of positioning technologies.

III. Methodology

A. Requirements Gathering

In order to retrieve the correct layout of the Physical Sciences building, an email was sent to the University Planning and Maintenance Office (UPMO) to request the building's floor plans. A 2D layout of the building was provided, separated into three files, with each file mapping out an entire floor.

For the labeling of the rooms, a physical visit inside the building was conducted to cross-examine the accuracy of the floor plan against the actual rooms, including the name of each room. After development of the application, another email was sent to UPMO to request user testing. Administrators from their office were asked to participate in testing the layout builder feature of the application.

B. Analysis and Design

Figure 1 shows the use-case diagram for the indoor navigation web application. Two actor roles were identified: users and administrators, each with distinct capabilities within the system. Users were able to search for rooms, select an origin and destination, and view the generated navigation route. Administrators, in addition to having access to navigation features, were able to create, edit, and publish floor layouts through the layout builder, giving them control over the data that powered the navigation feature for all users.

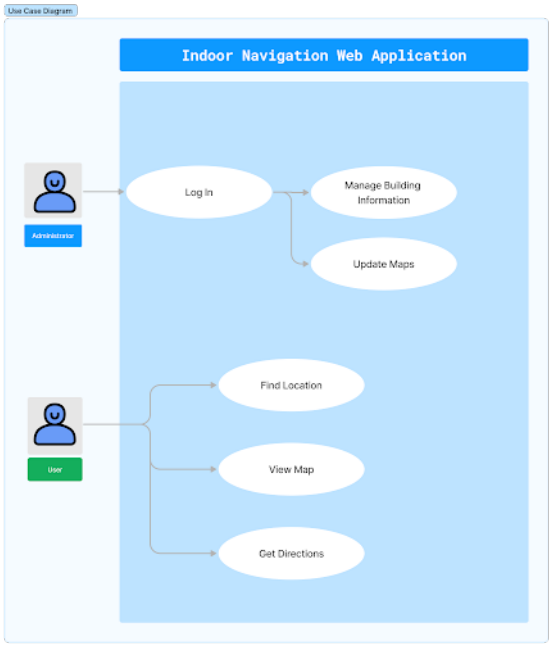


Fig. 1: Use-Case Diagram

Based on these use cases, the system was designed around a relational database structure using MariaDB. During backend startup, the database service connected to the database and verified that the required schemas were present before the application became available. A scope flag was incorporated within the relevant tables to separate live, published records from sandbox records, allowing administrators to edit and test layout changes without affecting the version currently used for navigation.

Figure 2 shows the Entity Relationship Diagram, which illustrates the database structure and the relationships between entities such as buildings, floors, rooms, and their associated navigation data.

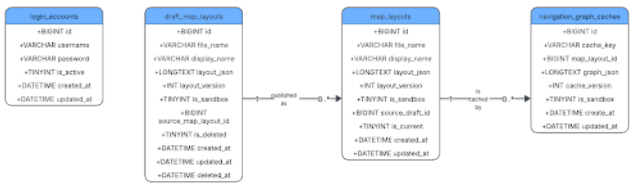


Fig. 2: Entity Relationship Diagram

C. Implementation

A laptop with the following specifications was used to develop the indoor navigation web application:

- Operating System: Windows 11 64-bit
- Processor: Intel® Core™ i5-8250U Processor 1.6 GHz
- Graphics Card: NVIDIA GeForce GTX 1050
- Memory: 20 GB DDR4 2800 MHz

The following software development tools will be used for the development of the application:

- Python: Python is an interpreted, object-oriented, high-level programming language used to build the backend component of the application.
- MariaDB: an open-source relational database management system.
- React: an open-source front-end JavaScript library used to build the frontend component of the application.
- Visual Studio Code: a source-code editor used for developing the application.

The application was built first with the layout builder features in mind. The goal was to create a working map layout with the integration of path routing and a persistence state to save layouts.

A layout of the Physical Sciences building was then created using the layout builder, guided by the floor plan provided by UPMO.

After that, focus shifted to the user interface, allowing users to select between different rooms.

Finally, once the application was built, deployment was carried out on a hosting service to test the application's stability outside the local development environment.

D. Deployment

The application was deployed through a mix of AWS services for the backend and database layer, with Render used to host the deployed frontend application.

The following AWS services were used:

- Elastic Compute Cloud (EC2) – an EC2 instance was created to configure and run the backend service
- Relational Database Service (RDS) – a managed cloud database that handled the storage of data and database operations

The application's backend and frontend were each packaged as separate Docker images. This containerization approach supported a more consistent and repeatable deployment process across the development lifecycle, since the same image built and tested locally could be deployed to the cloud without modification. The backend Docker image was deployed to an AWS EC2 instance, which handled the runtime of the backend service and processed incoming API requests. This backend container was configured to connect to an AWS RDS instance, which was responsible for persisting and retrieving application data. The frontend Docker image was deployed through Render, which built and hosted the containerized frontend application and made it publicly accessible via a web URL.

E. User Testing

User testing was conducted with two participant groups: regular users and building administrators. Building administrators tested the map layout builder by performing editing functions such as adding, modifying, saving, and publishing map elements, then verified that changes were reflected in the navigation interface. Regular users tested navigation accuracy to evaluate whether the application correctly guided them from a selected origin to a destination within the building.

A live demonstration of the application was conducted online via Zoom, with separate sessions held for regular users and building administrators. Questions related to application

features were accommodated during this period, and respondents were then asked to answer a Google Form containing questions related to system usability and navigation accuracy. A one-week testing period was given so that all users would have ample time to explore and use the features of the application.

IV. Results and Discussion

A. Application Features

1) Location Search: The application enables users to search for specific destinations within the building by providing an origin and a destination. This feature serves as the primary entry point for route generation. Figure 3 shows the location search feature



Fig. 3: Location Search

2) Multiple Floor Support: The application supports the viewing of buildings with multiple floors and allows users to navigate between the different floor levels, and provides the option to select different wings for each floor. Figure 4 displays the options for the user to select from specific areas of the building

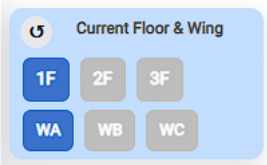


Fig. 4: Floor and Wing Selector

3) Map Visualization: The application provides an interactive indoor map that allows users to zoom, pan, and reset the viewing area. Figure 5 and 6 show the 2D floor view and 3D visualization when navigating

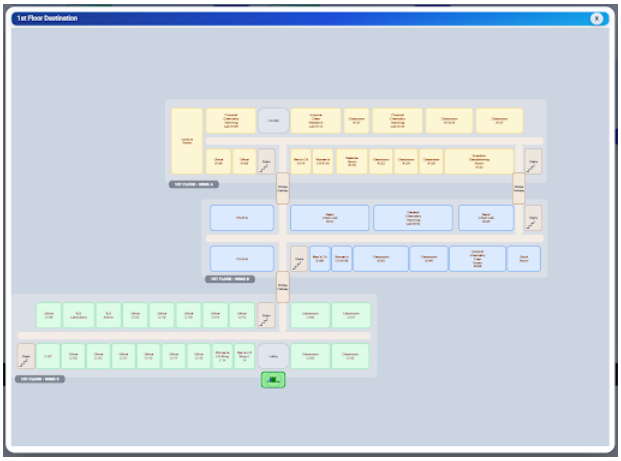


Fig. 5: 2D Map View

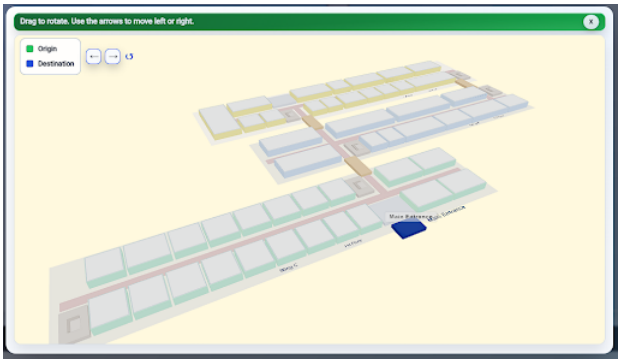


Fig. 6: 3D Map View

4) Multi-Floor Layout Builder: The application includes a layout builder that enables administrators to create and edit an indoor map to be seen by the users. Figure 7 shows the panel for the layout builder, which includes the options available for the administrator



Fig. 7: Layout Builder Panel

5) Layout Builder Map Drafts: The application allows administrators to save, load, rename, and delete draft map versions. Figure 8 and 9 show the Map Layouts and Map Drafts modals to allow the administrator to select between map versions

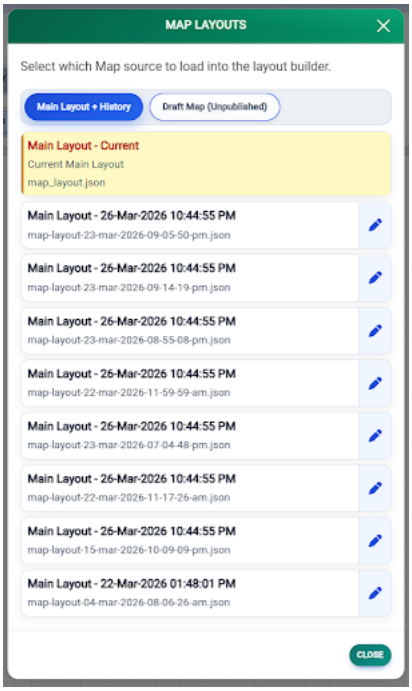


Fig. 8: Main Map Layouts

6) Sandbox Mode: The application includes a sandbox mode in which administrators can explore the features of the map layout builder without the risk of overwriting the main map edits and the published map being displayed to the users

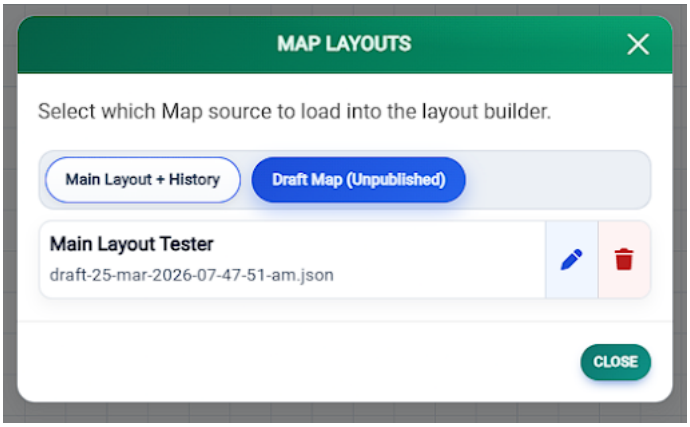


Fig. 9: Map Draft Layouts

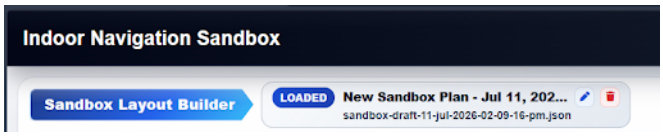


Fig. 10: Sandbox Mode

7) Auto-Generated Layouts: The application has an auto-generate map feature that allows the administrator to select a set of predefined layouts to assist them with map creation

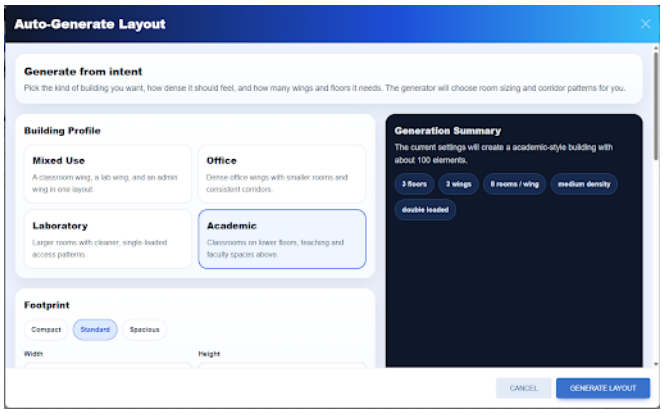


Fig. 11: Auto-generate Map Layout

B. Types of Users

1) Admin: Administrators can modify the mapping of the building, manage location data, and update floor plans.

2) User: Users can search for locations within the building and use the navigation features to access floors, rooms, and routes.

C. Mapping of the Floor Layout

The floor layout of the building is developed through the application's layout builder, where the admin creates and organizes the map structure for each floor. The admin directly encodes the building layout within the web application.

Using the layout builder, the admin defines the major spaces of the building, such as rooms, hallways, stairs, bridges, and entrances. In this process, each floor is treated as a two-dimensional plane with x and y coordinates, allowing the admin to position each component based on its x and y location and to size it according to its width and height within the floor plane. Labels are assigned so that the digital floor plan reflects the actual arrangement of the building.

The admin can revise the floor plan during the mapping process, and draft versions of the layout can be created and edited before publication. Once the admin finalizes the layout, the floor plan is published and becomes the reference used for navigation display within the application. Previously published versions of a map can also be reverted to and saved as a new draft, allowing the admin to roll back changes when needed.

After publishing the map, the application uses the mapped structure and computes the navigation routing based on the details provided by each component to determine the path from one point to another.

D. Navigation User Interface

Users can find specific locations within the building with search functionality and select their origin and destination points by typing in the search bars or by choosing the rooms in the room list selector. Figures 12 and 13 show the user interface from the user's perspective, whereas Figure 14 displays the interface from the administrator's perspective.



Fig. 12: Main User Interface

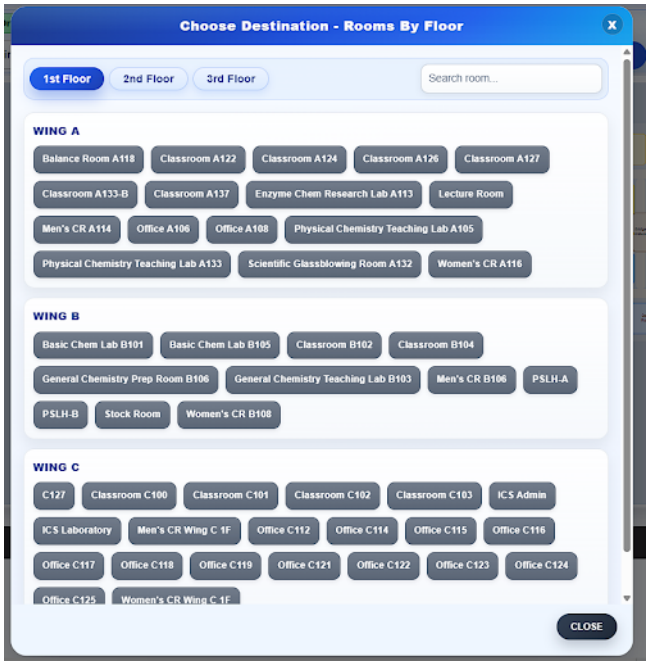


Fig. 13: Room Selector

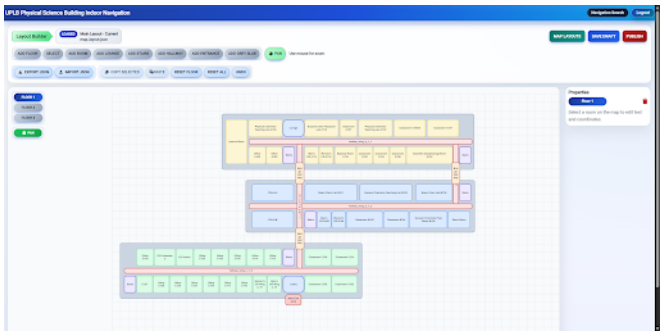


Fig. 14: Administrator User Interface

E. Non-Functional Requirements

- Usability: This requirement was met through a guided user interface. The application provides assisted search suggestions, room-list browsing, clear map controls, visible route overlays, and consistent navigation elements, making it easier for users to understand and operate the system.
- Performance: This requirement was met by having the application implemented to be browser-based, which means it can operate and is compatible on devices capable of running modern web browsers without requiring a platform-specific installation.
- Scalability – This requirement was met through the separation of frontend and backend services, containerized deployment, graph caching, and a database for storage persistence to accommodate for larger map layouts and multiple maps.
- Security – This requirement was met through administrator login, protected access to the layout builder only for administrators, session expiration, and controlled Cross-Origin Resource Sharing (CORS) settings. However, a more robust authorization would further improve security.
- Availability – This requirement was supported through the application's AWS-based deployment and service health mechanisms. The system was deployed on an AWS EC2 instance using Docker containers for both the frontend and backend, allowing the services to remain isolated and easier to restart when needed.

F. Testing

To evaluate the effectiveness of the developed web application in meeting its usability objectives, user testing was conducted with two groups of participants: regular users (students) and building administrators. This testing was intended to assess how effectively the system performed in actual use.

For building administrators, the focus of their testing was on the map layout builder feature, specifically whether the tools for the map layouts functioned correctly and were practical to use. They logged in to the system, accessed the layout builder, and tested the available editing functions such as adding, modifying, saving, and publishing map elements. They also verified whether the published changes were correctly reflected in the navigation interface.

For regular users, the focus of their testing was on navigation accuracy, particularly whether the application could correctly guide them from a selected origin to a specified destination within the building and whether the displayed route matched the actual path.

To evaluate the navigation accuracy, a qualitative testing phase was conducted. After completing the assigned navigation tasks, the participants were asked to answer a set of open-ended questions based on their experience using the application. The responses gathered from these questions were analyzed to determine whether the application was capable of providing accurate and understandable indoor navigation guidance.

To quantify the usability of the web application, the System Usability Scale (SUS) was used. This is a reliable tool for measuring the usability of various systems and services. The SUS provides a set of 10 questions with a five-point Likert scale, ranging from "Strongly agree" to "Strongly disagree." This scale is designed to evaluate a broad range of interactive products by deriving a single usability score that reflects a composite measure of the overall effectiveness, efficiency, and satisfaction with the system.

System Usability Scale Statements:

1) I think that I would like to use this system frequently.
2) I found the system unnecessarily complex.
3) I thought the system was easy to use.
4) I think that I would need the support of a technical person to be able to use this system.
5) I found the various functions in this system were well integrated.
6) I thought there was too much inconsistency in this system.
7) I would imagine that most people would learn to use this system very quickly.

8) I found the system very cumbersome to use.
9) I felt very confident using the system.
10) I needed to learn a lot of things before I could get going with this system.

Navigation Accuracy:

1) Did the application lead you to the correct destination? Describe what happened during the navigation process.
2) Was the route provided by the application accurate? Explain why or why not.
3) How easy was it to understand the route displayed on the map?
4) Were there any parts of the route that were confusing, unclear, or incorrect? If yes, describe them
5) Did you encounter any mismatch between the route shown by the application and the actual path that should be taken inside the building? If yes, describe the mismatch.
6) What suggestions can you provide to improve the navigation accuracy of the application?

For the building administrators, the following open-ended questions were asked:

1) Which parts of the layout builder felt easy to understand, and which parts felt unnecessarily complex?
2) Can you describe your experience creating or editing a building layout using the system?
3) How well did the different features of the layout builder work together while you were making changes?
4) Were there any parts of the layout builder that felt confusing, or difficult to predict? Please explain.
5) How confident did you feel that the layout you created or edited was correct and usable for navigation?
6) Would you like to use this application in the future?
7) What suggestions do you have to improve the layout builder so administrators can create more accurate and reliable maps?

A total of ten (10) responses were collected, with four coming from building administrators from the University Planning and Maintenance Office (UPMO) and six from students studying at the University of the Philippines Los Baños.

Table I shows the raw questionnaire results from the users. Table II shows the results of the SUS questionnaire for each respondent and the rating in which the score belongs. The six end users who completed the system usability questionnaire produced individual scores of 72.5, 85.0, 67.5, 97.5, 100.0, and 67.5, for a mean of 81.67 and a median of 78.75. The mean score of 81.67 falls within the "Excellent" range on the interpretative scale of SUS results, indicating strong overall satisfaction with the navigation experience. Two respondents scored at the lower end (67.5), suggesting that while the majority found the system highly usable, a subset encountered enough friction to rate the experience more moderately.

TABLE I: Raw Scores per Respondent

Respondent	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
R1	4	1	5	1	5	3	5	5	5	5
R2	3	1	5	1	4	3	4	1	5	1
R3	4	2	4	2	5	3	5	4	4	4
R4	5	1	5	1	5	1	5	1	5	2
R5	5	1	5	1	5	1	5	1	5	1
R6	3	1	4	3	4	1	4	2	4	5

TABLE II: Total Scores by Respondent and Score Rating

Respondent	SUS Score	Rating
R1	72.5	Good
R2	85.0	Excellent
R3	67.5	OK
R4	97.5	Excellent
R5	100	Excellent
R6	67.5	OK

Qualitatively, all six student respondents confirmed that the application successfully led them to their intended destination and described the routes as accurate or mostly accurate. Two respondents noted that routes involving stairways were occasionally not the shortest or most efficient option available, and another respondent recommended that the system be tested against a wider variety of stairway configurations. Respondents highlighted the three-dimensional view as an effective way to visualize paths across multiple floors compared to the two-dimensional view. Clarity in routing was rated favorably, with respondents describing the interface as straightforward and simple to understand.

The four building administrators who tested the layout builder feature described the tool as generally intuitive and responsive, useful for early-stage planning of floor layouts. Respondents pointed out that the interface was straightforward to navigate and use, and the resulting layouts helped communicate proposed arrangements to clients. One respondent noted that the application functioned effectively like a bubble diagram, a tool used for visualizing room connections before a final layout is committed. One feature identified for improvement was the addition of an autosave function; respondents mentioned that accidental misclicks could cause unsaved progress to be lost. Administrator respondents also pointed out that they had difficulty distinguishing the purpose of the Layout Builder from the Sandbox mode, which added complexity to understanding the separation between the two functions.

V. Conclusion and Future Work

A. Conclusion

Taken together, the two testing groups suggested that the application performed strongly on its core function, indoor navigation, while leaving room for refinement in the layout builder. Users rated the navigation experience highly, with a mean score falling within the "Excellent" range. Building administrators, on the other hand, found the layout builder and sandbox mode useful for building layout planning, helping communicate layout proposals to stakeholders and clients.

Some limitations of the application evaluation include the small sample size of both testing groups, consisting of six end users and four administrators. This limits how confidently these results can be generalized to a wider range of users. Additionally, testing was conducted during a single session for both types of participants, which does not capture how usability holds up with repeated or long-term use of the application. This is particularly relevant for building administrators, who may create and manage large, multi-story buildings over time.

B. Future Work

For future work, a primary area of focus is expanding the system beyond its current scope to cover other buildings across the campus, allowing users to navigate between buildings rather than within a single building alone. To complement this expansion, the development of native iOS and Android applications is recommended, as a transition from a web-based platform to dedicated mobile applications could offer an improved overall user experience.

The application should also integrate accessibility features, which involve mapping and displaying accessible routes, such as paths incorporating PWD (Persons with Disability) ramps, elevators, and accessible restrooms. Relatedly, integrating clear displays of emergency exit routes is recommended to enhance the safety and preparedness of all users within the building.

With respect to the navigation feature specifically, future versions should support real-time location tracking to guide users continuously as they move toward their destination, rather than relying on a static route display, and display doorways along the suggested path to further improve route clarity. Further routing optimization and testing are also recommended to broaden the scope of navigation scenarios evaluated, including more varied stairway configurations and larger, multi-floor paths.

On the administrator side, the layout builder would benefit from several refinements identified during testing. These include an autosave feature to prevent the loss of in-progress map drafts, a clearer distinction between the Layout Builder and Sandbox features to reduce confusion between the two, and support for real-world unit scaling with gridlines or rulers to improve drawing precision. The ability to upload a 2D or CAD reference plan to serve as a guide when constructing more accurate layouts is also recommended. Improving the responsiveness and synchronization of the layout builder's editing tools would further support administrators working on larger, more complex building layouts.

To support long-term map accuracy, a crowdsourced update and feedback system is recommended, through which users can report inaccuracies or suggest corrections directly within the application, helping keep building layouts current without relying solely on administrator updates.

Finally, integrating the system with outdoor campus navigation is recommended to provide a connected wayfinding experience for users moving between buildings.

ACKNOWLEDGMENT

I would like to express my deep gratitude to my adviser, Prof. Fermin Roberto G. Lapitan, for his thorough guidance and patience with me throughout this long journey toward completing this research. I am also grateful to my colleagues for their support as I balanced my work responsibilities and academic commitments. Finally, I am immensely grateful to my family for their constant love, encouragement, and unwavering belief that I will be able to finish my studies, which has continuously pushed me forward. I would never have reached this point without all of them.

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